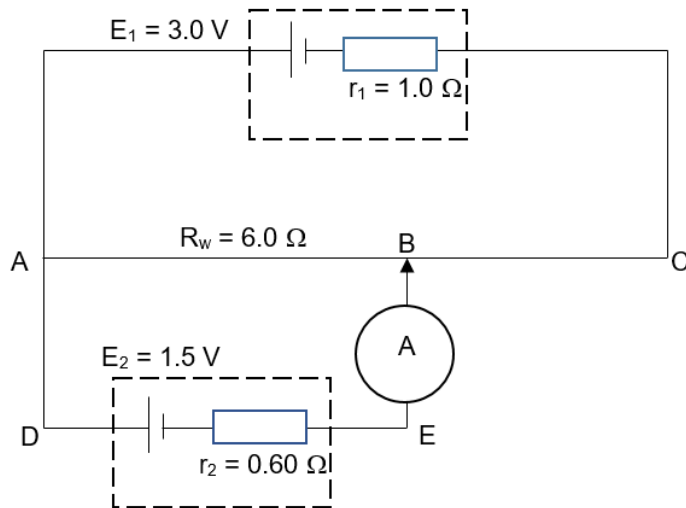


22. Ans: D



At balance, $V_{AB} = V_{DE}$

Since no current flows through the bottom cell at balance, $V_{DE} = E_2$

$$\text{Also, } \frac{L_{AB}}{L_{AC}} = \frac{V_{AB}}{V_{AC}} = \frac{V_{DE}}{V_{AC}} = \frac{E_2}{V_{AC}}$$

Then, $V_{AC} = E_1 \frac{R_w}{R_w + r_1}$ (apply potential divider principle to the top loop)

$$= 3.0 \times \frac{6.0}{6.0 + 1.0}$$

$$= 2.571 \text{ V}$$

So balance length, $L_{AB} = L_{AC} \frac{E_2}{V_{AC}}$

$$= 1.0 \times \frac{1.5}{2.571}$$

$$= 0.58 \text{ m}$$